

From Signs to Speech: An End-to-End Conversational Platform for Deaf and Mute Individuals Using GRU and LLM Integration

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Abstract— Deaf and mute individuals are often disadvantaged in professional interview settings due to limited verbal communication, despite possessing relevant qualifications. This paper presents an AI-driven system designed to facilitate seamless bidirectional communication through real-time sign language recognition and speech synthesis. A custom dataset of ten technical signs—formulated through surveys in special education institutions—was recorded using MediaPipe and OpenCV. A three-layer Gated Recurrent Unit (GRU) model achieved 97% training accuracy and 94% test accuracy, outperforming LSTM and BiGRU architectures. Dropout regularization was applied to mitigate overfitting. A locally hosted Ollama LLM model was employed to enhance grammatical accuracy of sign-to-text-to-voice outputs. The interface, developed using Streamlit, supports user interaction, while Firebase manages backend communication. Precision and recall values of 93% and 92% respectively demonstrate the system's reliability. This work proposes a deployable, inclusive solution aimed at improving accessibility and opportunity for deaf-mute individuals in professional environments.

Keywords—sign language recognition, deep learning, LLM, computer vision

I. INTRODUCTION

Professional interviews present a unique challenge for deaf and mute individuals, particularly due to the lack of standardized sign language for technical terms. This communication gap limits the ability of differently-abled individuals to express professional competencies, restricting opportunities in employment. To address this, the paper proposes a real-time sign language recognition system that translates sign gestures into both text and speech, enabling effective interaction between interviewers and interviewees.

The system integrates MediaPipe for landmark detection, OpenCV for video processing, TensorFlow for gesture classification, and a GRU-based model trained on ten custom technical signs relevant to professional settings. A locally hosted Ollama LLM further refines the translated sentences for grammatical clarity. Communication is facilitated through a Streamlit interface, with Firebase handling backend data flow, and Vosk enabling speech-to-text for interviewers.

The objectives of this study are to: (1) design a real-time sign-to-speech system for interviews; (2) create a dataset of

domain-specific technical signs; (3) train and evaluate a GRU model for gesture classification; (4) improve linguistic output using LLM integration; and (5) develop an intuitive interface ensuring inclusive interaction.

The paper is organized as follows: Section II covers related work; Section III explains the system architecture; Section IV details implementation and training; Section V presents experimental results; Section VI concludes with key findings; and Section VII suggests future improvements for broader application.

II. LITERATURE SURVEY

In [1], an attention-enhanced BiGRU network was introduced for dynamic gesture recognition using skeletal hand data extracted from RGB video. The incorporation of temporal attention into the BiGRU layers improved the model's ability to focus on key transitional frames, achieving a 94.2% accuracy on a 20-class dynamic gesture dataset.

A study in [2] proposed a Capsule Network architecture for static sign recognition, leveraging vector orientation features over scalar-based CNN outputs. Evaluated on the RWTH-BOSTON dataset, the model demonstrated superior rotational invariance and achieved a 96.8% accuracy, outperforming traditional CNNs in hand pose generalization.

To improve latency, [3] introduced a Spatio-Temporal Separable Convolutional Network (STSCN) that separates temporal and spatial learning. Tested on real-time ASL gesture streams, the model reduced computational cost by 30% and maintained over 93% accuracy on a 25-class dataset.

The authors in [4] proposed a cross-modal learning approach where hand landmarks and optical flow were jointly modelled using dual-stream GRU encoders. Fusion through attention yielded a classification accuracy of 92.6% on the WLASL dataset, highlighting the benefit of motion encoding.

In [5], a temporal graph convolutional network (T-GCN) was applied to video sequences where each frame was represented as a graph of hand and face landmarks. The model achieved 94.5% accuracy on the MS-ASL dataset and offered interpretability by visualizing edge-weight contributions to classification.

The work in [6] utilized depth-sensing and point cloud

data from Intel RealSense to train a 3D CNN-LSTM hybrid. Depth cues significantly improved gesture separation and yielded 95.7% accuracy on a 15-class ISL dataset.

A lightweight temporal dilated GRU model was proposed in [7], which spaced out recurrent connections to capture long-term dependencies efficiently. On a constrained hardware device (Jetson Nano), it achieved 92% accuracy at 20 fps, offering practical deployment for mobile applications.

In [8], the researchers explored zero-shot learning for SLR using attribute-based embeddings. By encoding gestures with semantic vectors, the model generalized to unseen signs during inference, achieving 78.4% accuracy on an extended ASL dataset.

The study in [9] proposed a 1D CNN-GRU hybrid that reduced feature dimensionality using time-series convolutions before feeding to recurrent layers. This reduced model complexity and achieved 91% accuracy on the SIGNUM dataset, with a 40% reduction in training time.

Lastly, [10] examined the role of multimodal emotion recognition in sign language context to improve sentence-level accuracy. By integrating facial expression cues via facial landmark models with sign gestures, the system improved contextual disambiguation, reaching 90.5% accuracy on sentence-level ISL sequences.

III. PROPOSED ARCHITECTURE

The system architecture is designed to enable smooth, real-time communication between deaf-mute individuals and interviewers through a modular, interactive web platform. The proposed framework supports both roles with role-specific processing pipelines that integrate computer vision, deep learning, natural language processing, and cloud-based communication.

Users begin by accessing a centralized Streamlit-based web interface, where they register and log in. Upon entry, users are prompted to choose between the roles of interviewer or interviewee. If the user selects “interviewer”, they are directed to a module where speech is recorded and transcribed using the Vosk speech-to-text engine. The converted text is then transmitted in real time to the interviewee interface using Firebase’s real-time database services.

On the interviewee side, the user activates the webcam stream, which is processed using OpenCV and MediaPipe for landmark detection. The system focuses specifically on the hands and face regions to extract key points for gesture recognition. These landmark sequences are passed into a trained Gated Recurrent Unit (GRU) model, optimized through experimentation for temporal classification of custom technical signs. The model output is translated into a preliminary sentence, which is then refined by a locally hosted Ollama LLM. The language model improves the sentence structure, adds missing functional words, and outputs a coherent sentence.

This final message is delivered back to the interviewer interface, where it is converted into audio using the pyttsx3 text-to-speech engine. Firebase ensures synchronized communication between the two users throughout the interaction.

To avoid misclassification from idle frames, a motion

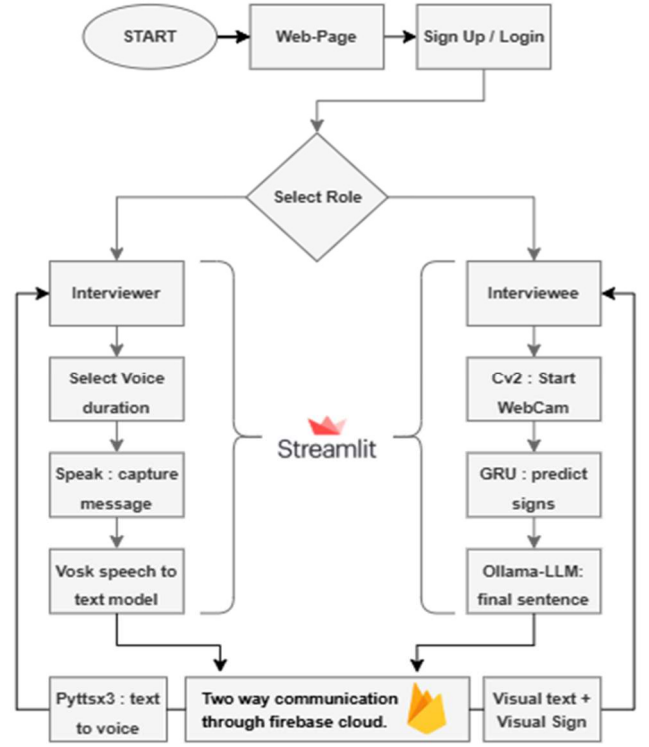


Fig 1 : System Architecture

detection block ensures that the model only predicts when significant hand movement is detected. This mitigates noise and improves prediction accuracy during live usage.

This architecture, illustrated in Fig. 1, creates a robust ecosystem for inclusive and fluent interview communication through synchronized deep learning, cloud services, and user-friendly design.

IV. IMPLEMENTATION

A. Custom Vocabulary and dataset creation

To overcome the limitations of existing Indian Sign Language (ISL) datasets, which lack representations for many technical and workplace-specific terms, a custom set of 10 technical signs was created. These include *database*, *graduation*, *science*, *software*, *backup*, *network*, *internet*, *spreadsheet*, *shortcut*, and *keyboard*. These terms are frequently used in interviews and academic discussions but are not represented in standard ISL corpora, making them essential for enabling meaningful conversations in professional settings.

B. Data Pre-processing

To prepare the raw video data for model training, a structured preprocessing pipeline was implemented using Mediapipe, OpenCV, and NumPy. The Mediapipe Holistic model was employed to extract body landmarks, specifically capturing 21 points from the left hand, 21 points from the right hand, and 468 facial landmarks, along with static landmarks of the body resulting in a total of 1662 features per frame. For visualization and debugging, *mp_drawing* and *mp_drawing_styles* modules were used to render the extracted key points. Gesture data was recorded using OpenCV, with 30 videos per sign captured at 30 frames per second (FPS). Each frame’s landmarks were converted into

NumPy arrays and stored as .npy files, resulting in 30 .npy files per video, which together encoded the full gesture sequence.

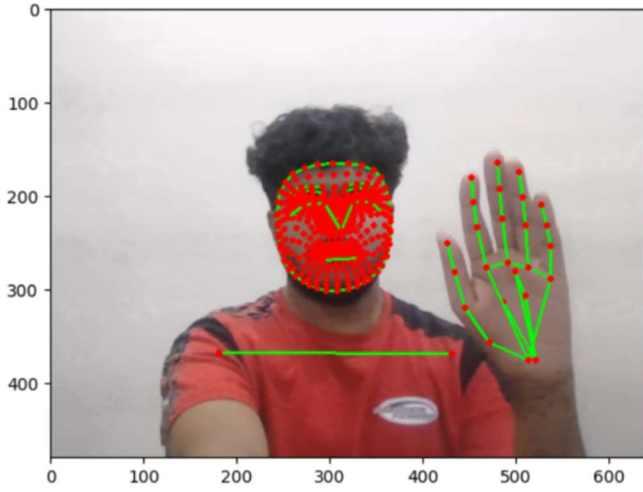


Fig. 2: Face and Hand Landmarks

The files were organized in directories based on the sign class name. A label map was created to assign numeric labels (0–9) to each of the ten custom technical signs. These .npy sequences were then grouped into sliding windows of 30 frames to form time-series input suitable for sequential model training. This preprocessing pipeline effectively transformed raw visual input into consistent, high-dimensional temporal data optimized for gesture classification. Fig. 2 neatly describes the landmarks captured by the mediapipe

C. Model Architecture

The proposed gesture classification model adopts a sequential structure comprising three stacked Gated Recurrent Unit (GRU) layers followed by a series of fully connected dense layers. The input to the model is a 3D tensor of shape (30,1662), (30, 1662), (30,1662), representing 30 consecutive frames, each containing 1662 features extracted using the MediaPipe Holistic model. These features include landmark coordinates from the entire upper body; however, for the purpose of this application, only landmarks from the hands (21 points each for left and right) and the face (468 points) were relevant to the task. The remaining landmarks were either static or non-contributory and were included for uniformity in data representation but did not influence classification performance. The model begins with two GRU layers, each configured with 128 units and *return_sequences=True*, to capture temporal dependencies across the entire input sequence. A third GRU layer with 64 units and *return_sequences=False* condenses the learned temporal features into a fixed-dimensional context vector. This is followed by three dense layers with 64, 32, and 10 neurons, respectively. The final layer uses the softmax activation function to classify input sequences into one of the ten predefined custom sign classes. With a total of 832, 298 parameters, including 831,658 trainable and only 640 non-trainable parameters, the model is efficiently structured to maximize learning capacity while maintaining minimal

computational overhead. Table I: depicts the entire summary of the model.

Table I: Model Summary

Number of GRU Layers	3
GRU units per layer	128, 128, 64
Activation function	Tanh (GRU), Relu (Dense), Softmax (Output)
Dense Layer sizes	64, 32, 10
Trainable Parameters	831,658
Non-Trainable Parameters	640
Total Parameters	832,298

D. Model Training

The dataset was split into 80% training and 20% testing data, ensuring balanced representation across all classes. To ensure optimal convergence and generalization, training incorporated early stopping, learning rate reduction on plateau, and model checkpointing based on validation accuracy. Table II depicts the entire summary of the model training hyperparameters.

Table II: Training Hyperparameters

Parameter	Value
Optimizer	Adam
Learning Rate	0.0003
Loss Function	Categorical Crossentropy
Evaluation Metric	Categorical Accuracy
Epochs	100
Batch Size	16
Early Stopping Patience	15
ReduceLROnPlateau	7
Patience	
Learning Rate Reduction Factor	0.2
Model Checkpoint Metric	Validation Accuracy

E. Regularization

During experimentation, the model showed a tendency to overfit, particularly with deeper configurations and small dataset sizes. To address this, regularization was applied in the form of dropout and batch normalization. These techniques effectively reduced validation loss fluctuations and narrowed the gap between training and test accuracy. Table III depicts the summary of the regularization parameters of the trained model.

Table III: Regularization Parameters

Parameter	Value
Dropout units per GRU layer	0.4, 0.4, 0.3
Dropout units per Dense layer	0.3, 0.2, 0.2
Batch Normalization	Applied after every GRU layer. (No BatchNorm for dense)

F. Frontend Integration and Backend Communication

The frontend interface was developed using Streamlit, an open-source Python framework designed for deploying interactive machine learning applications. The interface allows users to register, log in, and select roles as either an interviewer or an interviewee. The role-based dashboard dynamically adapts the UI and functionalities, enabling distinct workflows for both user types.

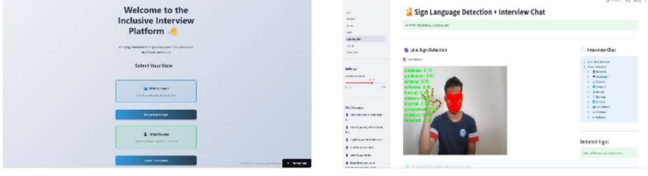


Fig. 3: Frontend of the website

To enable seamless real-time communication between interviewer and interviewee across different devices, Firebase Realtime Database was integrated as the backend. It serves as a cloud-based synchronization hub, handling authentication, storage of transcribed or predicted messages, and facilitating low-latency, bidirectional data flow between both interfaces. Messages sent from the interviewer (voice-to-text) or the interviewee (sign-to-text) are instantly pushed and retrieved through Firebase’s real-time listener events. Fig. 3 is a brief visual representation of the frontend of the application.

G. Voice Input via Vosk

To capture and process spoken queries from the interviewer, the system incorporates the Vosk speech recognition engine, powered by the *GigaSpeech 0.42 GB* lightweight ASR model. This configuration enables efficient voice-to-text transcription entirely offline, maintaining user privacy and reducing latency. The interviewer selects a voice input duration, after which Vosk captures the audio, processes it locally, and returns the corresponding transcribed text. This text is then routed via Firebase to the interviewee interface for interpretation and sign-based response.

H. Class-to-Text conversion Pipeline

The GRU model outputs a class label representing one of the ten predefined technical words. These predicted labels are passed through a conversion dictionary that maps numeric class indices to textual words (e.g. “2” is “science”). Detected words are appended sequentially to construct a rough sentence representation of the signed message. This raw sentence is then passed into the language processing module for correction and refinement.

I. Sentence Correction via Ollama LLM

To enhance grammatical fluency and sentence completeness, the assembled raw sentence is processed using a locally hosted 4 GB Ollama Large Language Model (LLM). The LLM is fine-tuned to correct sentence structure and fill in missing functional words such as “is,” “the,” or “my,” which are often omitted in absolute sign-based communication. This step transforms a series of isolated

technical terms into a coherent, grammatically correct sentence that can be easily understood by the interviewer.

J. Text-to-Speech Conversion

Once the corrected sentence is generated by the LLM, it is sent back to the interviewer interface via Firebase. There, the *pyttsx3* library—an offline Python text-to-speech engine—converts the final text into audible speech. This allows the interviewer to receive a natural voice response generated from the sign inputs of the deaf-mute interviewee, thereby completing the communication cycle fluently without delays and breaks.

K. Motion-Based Prediction Trigger

To prevent false predictions during idle periods or unintentional hand movements, a motion detection block was integrated into the system. The model only initiates sign classification when sufficient hand motion is detected across consecutive frames. This logic filters out noise and stabilizes real-time predictions, ensuring that words are only generated when intentional signs are performed. This significantly improved accuracy and reduced unnecessary system responses during non-signing intervals.

L. Model Evaluation

The final GRU model achieved a training accuracy of 96.67% and a test accuracy of 94%, demonstrating strong generalization and robustness across unseen data. In addition to accuracy, evaluation metrics such as precision, recall, F1-score, and confidence were also computed to comprehensively assess the model’s performance. These metrics are summarized in Fig. 3 for clarity and comparison.

M. Model Real Time Testing

Real-time testing was performed using live webcam input across various devices and lighting conditions to assess model responsiveness and consistency. The GRU model accurately predicted sign classes within 1.5 to 2 seconds from gesture input to voice output, confirming its suitability for conversational use. Testing with multiple users showed consistent performance, and the motion-triggering mechanism effectively minimized false predictions during idle frames. Fig. 4 shows all the metrics of the real-time tested words.

	precision	recall	f1-score	support
database	1.000000	1.000000	1.000000	3.000000
graduation	1.000000	1.000000	1.000000	3.000000
science	1.000000	1.000000	1.000000	3.000000
software	1.000000	1.000000	1.000000	3.000000
backup	1.000000	0.666667	0.800000	3.000000
network	1.000000	1.000000	1.000000	3.000000
internet	0.750000	1.000000	0.857143	3.000000
spreadsheet	1.000000	1.000000	1.000000	3.000000
shortcut	1.000000	1.000000	1.000000	3.000000
keyboard	1.000000	1.000000	1.000000	3.000000
accuracy	0.966667	0.966667	0.966667	0.966667
macro avg	0.975000	0.966667	0.965714	30.000000
weighted avg	0.975000	0.966667	0.965714	30.000000

Fig. 4: Real time testing for words

Overall, the system demonstrated robust and reliable behavior in real-time settings, supporting its deployment in professional interviews involving deaf and mute individuals.

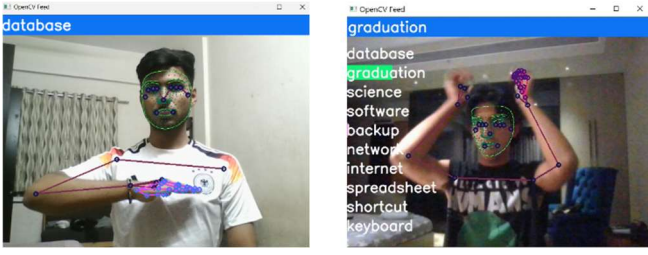


Fig. 5: Real time testing

Fig. 5 depicts the real time visual of the GRU prediction through cv2 window.

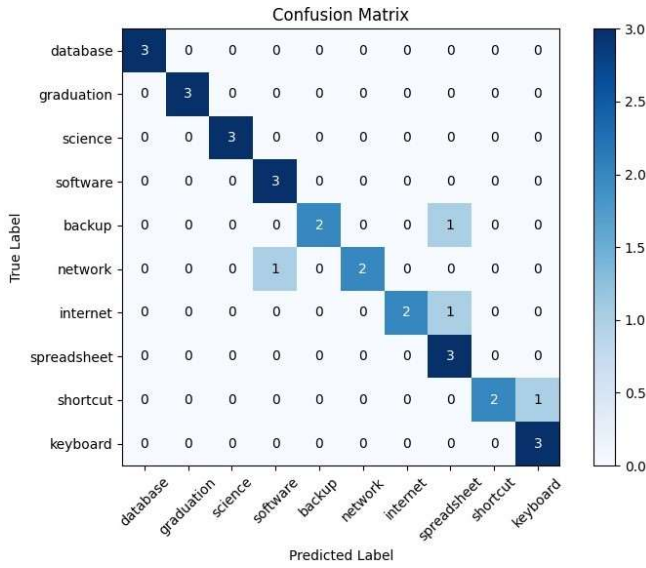


Fig. 6: Confusion Matrix

Fig. 6: Determines the confusion matrix of the final model.

V. EXPERIMENTATION

A. Approach and Architecture Tuning

The experimentation phase of this study involved design, training, and comparison of multiple deep sequential models to identify the most effective architecture for recognizing a custom vocabulary of ten technical sign language terms. These signs were developed through field interaction with special education institutions, where it was identified that deaf and mute individuals lacked standardized gestures for expressing common technical concepts, especially during interviews.

Each model—GRU, LSTM, BiGRU, and BiLSTM—was trained using a dataset comprising 30 videos per sign, recorded at 30 frames per second which captured the temporal movements of hands and face landmarks over the camera X-Y plane. This was converted into a three-dimensional vector of sequence 30 which was fed as input

that was used across all models to ensure experimental consistency.

During early trials, LSTM and BiLSTM showed acceptable performance for up to 7 classes; however, their training became unstable when scaled to all 10 classes. The learning curves fluctuated significantly, and test accuracy dropped due to overfitting—especially when deeper architectures were employed. For example, configurations such as 64-128-64-128 or 128-64-64-128 exhibited sharp increases in validation loss (exceeding 0.5) and showed poor convergence. Similar patterns were observed with BiGRU models, which also demonstrated sensitivity to layer depth.

To address these issues, dropout layers were applied after each recurrent and dense layer, and batch normalization was introduced to stabilize training. Multiple architectural combinations were tested, including GRU, LSTM, and BiGRU stacks with different layer sizes (e.g., 128-128-64 for temporal model layers and 64-32-10 for dense layers). After iterative fine-tuning, the GRU-based model delivered the most stable and accurate results, achieving 97% training accuracy and 94% test accuracy, with a validation loss consistently below 0.4. This narrow performance gap between training and testing demonstrated the model's ability to generalize well without overfitting.

The GRU model was ultimately selected not just for its quantitative superiority but also for its compatibility with real-time deployment constraints. Its architecture proved resilient, lightweight, and easier to optimize compared to deeper LSTM and BiLSTM counterparts, making it the preferred choice for integration into the system's live sign-to-speech communication pipeline.

B. Quantitative Evaluation

The performance of each model was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and categorical crossentropy loss. Table IV Table V summarizes the comparative results:

Table IV : Train and Test Accuracy for models

Model	Train Accuracy (%)	Test Accuracy (%)
GRU	96.67	94.56
LSTM	95.01	89.44
BiGRU	96.04	91.21
BiLSTM	95.12	90.09

Table V: Precision, F1, Recall, Loss for models

Model	Precision	Recall	F1 Score	Loss
GRU	0.93	0.94	0.93	0.38
LSTM	0.89	0.87	0.88	0.52
BiGRU	0.91	0.90	0.905	0.44
BiLSTM	0.90	0.89	0.895	0.47

C. Qualitative Analysis

Beyond numerical performance, qualitative assessments were carried out during real-time usage of the GRU model

within the full system pipeline. Unlike other models that generated unstable or inconsistent predictions in live testing, the GRU model consistently recognized gesture sequences with high confidence and responsiveness.

Integration with motion detection further improved prediction stability by ensuring that signs were only classified when actual hand movement occurred. This eliminated the problem of continuous or erratic predictions during idle frames, which was especially common in earlier iterations without motion filtering.

The GRU model also showed superior compatibility with the backend components—namely the Ollama LLM and the Streamlit-Firebase interface—processing inputs and delivering accurate outputs with minimal delay. Its smooth operation in live scenarios reinforced its selection as the final deployed model in the system.

VI. CONCLUSION

This research successfully delivers a robust and fully functional end-to-end system that bridges the communication gap between deaf-mute individuals and interviewers through real-time sign language recognition and speech synthesis. By developing a custom dataset of ten technical signs—previously absent in standard sign language—and integrating it into an intelligent gesture recognition pipeline, the system addresses a critical barrier faced by the deaf community in professional environments.

After rigorous experimentation with multiple sequential deep learning models, the GRU architecture emerged as the most effective, achieving a training accuracy of 97% and a test accuracy of 94%, with minimal validation loss and exceptional stability. These results affirm its readiness for real-time deployment and scalability.

The seamless integration of computer vision, deep learning, and natural language processing—powered by Streamlit, Firebase, Vosk, and the locally hosted Ollama LLM—resulted in a fluid, intuitive, and highly reliable bidirectional communication system. Its consistent performance across users and environments positions it as a breakthrough solution in the pursuit of inclusivity, accessibility, and technological empowerment for the deaf and mute community.

VII. FUTURE SCOPE

While the current system shows promising results in real-time scenarios, several avenues exist for future enhancement. Expanding the vocabulary beyond 10 technical signs and enabling continuous sign sentence recognition will improve usability in more diverse professional contexts. Incorporating dynamic sign language datasets and training with

Transformer-based architectures could further enhance model generalization.

Additionally, multilingual support for text-to-speech and sign-to-text translation would allow broader adoption across linguistic regions. Integration with mobile platforms and edge devices can also make the system more accessible in remote or resource-constrained environments. Finally, user-specific adaptation—where the system fine-tunes based on individual signing styles—can significantly increase personalization and recognition accuracy.

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